The Herschel Mirror as a Dish Antenna: An Archival Total-Power Search for a 541.7 GHz Line

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Abstract

A flat or curved conductor immersed in a kinetically mixed vector field of mass m radiates photons of energy mc^2. We used the 3.5 m primary mirror of the Herschel Space Observatory as such a dish antenna, with archival HIFI band-1a heterodyne data as the receiver.

Target. We searched at 541.664 GHz ($m = 2.240$ meV). This is the only vacuum-induced quantum mass of a gravity framework that ties the galactic acceleration scale to the dark-energy density.

The method. A mirror-generated line cancels in the archived (reference-subtracted) spectra, because both chop phases view the sky through the primary. We therefore rebuilt total-power spectra from raw Level-0 counts against Herschel's internal hot and cold loads. A real line must sit at a fixed spacecraft-frame frequency in every observation, whatever the target.

Result. From 49 observations (7.0 h; 150 spectra), we find no line: $a = -6.7 \pm 7.7$ mK (DSB peak), $z = -0.86$. The 95% upper limit is 12.7 mK. This gives $\varepsilon < 1.9 \times 10^{-11}$ for a dish-coupling factor $S = 50$ (range 8.6×10^{-12} to 8.6×10^{-11} for $S = 10$ to 1000). All four pre-registered controls pass, and a mutation test recovers an injected line at 9σ.

What it closes. The bound is direct and does not depend on cosmology. It closes the window in which this field could be converted to light at useful efficiency ($\varepsilon \approx 4$–6×10^{-9}) by a factor of ∼200. It does not reach the framework's falsification level.

1 Motivation

- **The framework.** The framework's galactic law is $g_N = g^2/(g + a_0)$, with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$. The coefficient $\frac{1}{2}$ is fitted. With this identification, the only quantum mass built from the vacuum is $m = (\rho_{\rm DE}\hbar^3/c^3)^{1/4} = 2.240$ meV, i.e. $\nu = 541.7$ GHz [1].
- **The framework's prediction.** Its extra gravitating component ("cold energy") has no non-gravitational channel, so no line should exist.
- **The alternative.** A hypothetical portal by which cold energy could be converted to light would require a kinetic mixing $\varepsilon \approx 4$–6×10^{-9} at this mass [1].
- **The proposal this follows.** A dedicated dish search at this frequency was proposed in PAPER52 [2]. This note asks what existing data already say.

2 Signal

- **Power.** A conductor of area A in a vector field of density ρ emits $P = \frac{2}{3}\varepsilon^2\rho c^3 A$ [3]. We take the framework's local density, $\rho = 0.01\, M_\odot\,{\rm pc}^{-3}$, and Herschel's primary mirror, $A = 9.6\,{\rm m}^2$.
- **Geometry factor S.** A parabolic telescope collects only part of this emission at its focus. The literature finds power losses of order tens for dish antennas [4]. We therefore report $S = 10$, 50 (primary) and 1000.
- **Line width.** The line is ∼0.3 MHz wide, set by the galactic velocity dispersion, which is narrower than a 1.1 MHz HIFI WBS channel.

3 Data and method

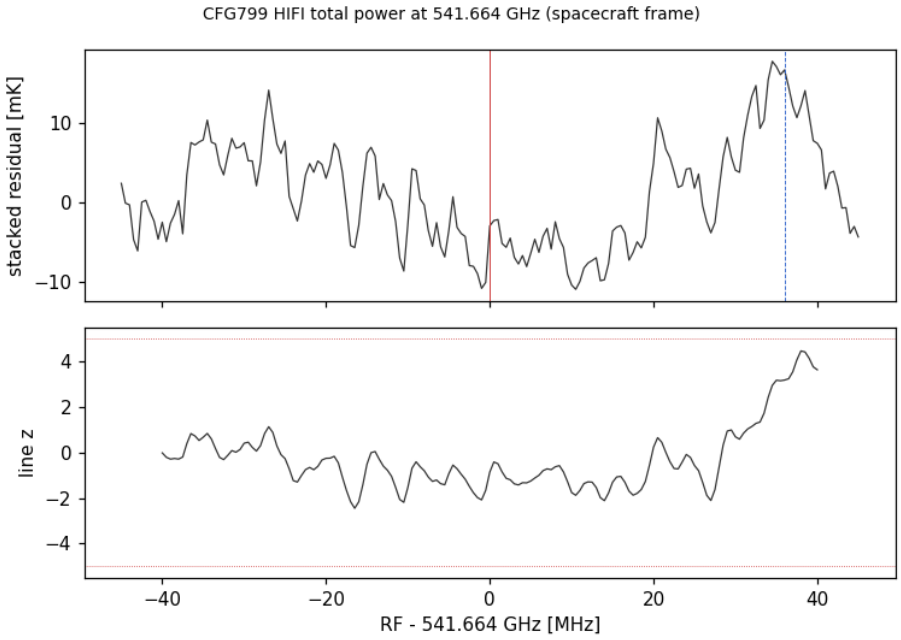
- **Sample.** 95 public band-1a observations put 541.7 GHz in a sideband (lane CFG798). We reduced 55: all 53 DBS/FastDBS point observations and 2 spectral scans. We did not use the rest:
  - 5 LoadChop/raster observations were excluded by the frozen mode rule;
  - 35 large spectral scans could not be downloaded because the archive server truncates ∼1 GB transfers;
  - 2 observations failed the Level-1 consistency anchor;
  - 4 lost all spectra to the masks.

  This leaves 49 observations (150 spectra; 7.0 h).

- **Total power.** For each sky phase we form $(C_{\rm sky} - C_{\rm cold})/(C_{\rm hot} - C_{\rm cold})$ from Level-0 counts. Frequency axes and a timing anchor come from Level-1. The internal loads never view the primary.
- **Statistic.** We fit a fixed-width Gaussian plus a degree-3 baseline within ±50 MHz in each spectrum, and take the weighted stack in spacecraft-frame RF across observations, LO settings and polarisations.
- **Masks.** We mask the 557 GHz water-line image (LO ≈ 549.3 GHz), sub-band edges, flagged channels, and astrophysical lines identified in Level-1. A source line appears in Level-1, while a mirror line would not.
- **Pre-registration.** All criteria were committed before any target statistic was computed: classes, masks, controls, and the coupling conversion. Two deviations to the timing anchor are disclosed in the lane README; both were made before the final run.

4 Results

	541.664 GHz (primary)	541.700 GHz (secondary)
stacked amplitude (DSB peak)	-6.7 ± 7.7 mK ($z = -0.86$)	$+20.5 \pm 6.5$ mK ($z = +3.17$)
Level-1 (reference-subtracted)	$z = -0.72$	$z = +0.03$
95% limit	12.7 mK	31.1 mK
ε_{95}, $S = 10/50/1000$	8.6×10^{-12} / $\mathbf{1.9 \times 10^{-11}}$ / 8.6×10^{-11}	1.3×10^{-11} / 3.0×10^{-11} / 1.3×10^{-10}



Stacked total-power residual (top) and line significance (bottom) around 541.664 GHz, in the spacecraft frame.

Controls (all pass).

- Injection recovery: exact. A mismatched profile recovers 0.94.
- Null offsets at ±5, 10 and 20 MHz: $p = 0.24$.
- An IF-shuffled stack over 3040 fits is consistent with zero ($z = +0.20$).
- The Level-1 anchor passes for 53 of 55 observations.
- A mutation run that injects a 77 mK line into total power only recovers it at $z = 9.1$ and classes it as a line candidate.

The 541.700 GHz excess. The 3.2σ excess at the secondary frequency is below threshold. It falls in the look-elsewhere region: the largest $|z|$ in a ±40 MHz scan is 4.45, near the same frequency. It is also within the Doppler range of source lines across epochs. We report it as a hint only. The undownloaded spectral scans could test it.

Per-observation converter test. A converter-strength line would be 28–2800 K (DSB peak), far above the noise of any single spectrum. All 48 observations tested individually exclude it at 5σ for every S.

5 Interpretation

- **The converter window is closed.** It needs $\varepsilon \approx 4$–6×10^{-9}, and we exclude $\varepsilon > 1.9 \times 10^{-11}$, a factor $\sim$200 ($\sim$47 even at $S = 1000$). CMB spectral-distortion bounds also exclude it ($\varepsilon \lesssim 10^{-13}$, provisional) [5,6], but those are indirect: they assume the field existed and coupled in the early universe. The Herschel bound measures the local field directly.
- **The framework survives.** It predicts no line. Its falsification level ($\varepsilon \sim 10^{-13}$, PAPER52 v1.1 [2]) is a factor of $\sim$190 below this bound.
- **The limiting factor.** The stack is limited by baseline structure ($\chi^2_\nu = 2.8$), not by receiver noise. A dedicated dish with a stable baseline remains the decisive experiment.
- **Novelty.** To our knowledge, this is the first use of archival space-telescope heterodyne data as a dish-antenna search, and the first direct search at 2.2 meV. Earlier dish searches covered 0.67–0.92 meV [4].

6 Caveats

- The coupling factor S, the effective sideband gain (taken as 0.5), the load coupling (taken as 1) and the local density are provisional. The limits scale with them.
- 4.2 h of spectral-scan data were not used.
- The target frequency relies on the fitted coefficient $\kappa = \frac{1}{2}$. The alternative footing gives 595.5 GHz, which has too little archival coverage here.

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[1] This programme (public repository): ARGON03, ARGON06; lanes CFG723, CFG795, CFG797, CFG798, CFG799 (frozen criteria, scripts, outputs). [2] PAPER52, "A Two-Frequency Sub-Millimetre Dish Search at 542 and 596 GHz", v1.1, doi:10.5281/zenodo.23404581. [3] D. Horns, J. Jaeckel, A. Lindner, A. Lobanov, J. Redondo, A. Ringwald, JCAP 04 (2013) 016. [4] S. Knirck et al., JCAP 11 (2018) 031. [5] A. Caputo, A. J. Millar, C. A. J. O'Hare, E. Vitagliano, PRD 104 (2021) 095029. [6] B. Cyr, J. Chluba et al., MNRAS 535 (2024) 1874. Herschel is an ESA space observatory with science instruments provided by European-led Principal Investigator consortia and with important participation from NASA. Data from the ESA Herschel Science Archive. AI assistance: Anthropic's Claude; parallel AI sessions.